Topic: Primer/probe design for qpcr

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Development of an Automated qPCR Primer and Probe Design Tool for Pathogen Detection and Quantification Using Python

Introduction

Quantitative Polymerase Chain Reaction (qPCR) is a widely used molecular biology technique for the detection and quantification of pathogens, including bacteria, viruses, fungi, and parasites. Due to its high sensitivity, specificity, and rapid turnaround time, qPCR has become an essential tool in clinical diagnostics, food safety testing, environmental monitoring, and infectious disease surveillance. The accuracy of pathogen detection depends largely on the design of specific primers and probes that target unique regions of a pathogen's genome.

Problem Statement

The design of qPCR primers and probes for pathogen detection requires careful consideration of several parameters, including melting temperature (Tm), GC content, amplicon size, and secondary structure formation. Manual design is often labor-intensive, time-consuming, and prone to human error. Additionally, many existing software tools are proprietary or offer limited customization. Therefore, there is a need for an accessible bioinformatics tool that can automate the design and evaluation of qPCR primers and probes for pathogen detection and quantification.

Rationale of the Study

Rapid and accurate pathogen detection is crucial for disease diagnosis, outbreak management, and public health surveillance. Automating primer and probe design can improve assay development efficiency, reduce design errors, and support the creation of highly specific diagnostic assays. This project also demonstrates the practical application of bioinformatics and computational biology in addressing challenges in molecular diagnostics and infectious disease research.

Aim

To develop a Python-based bioinformatics tool that automatically generates and evaluates qPCR primer, and probe sets for pathogen detection and quantification from DNA sequences.

Objectives

- Read pathogen DNA sequences from FASTA files.
- Generate candidate forward and reverse primers.
- Calculate GC content and melting temperature.
- Filter primers according to established qPCR design criteria.
- Design internal TaqMan probes targeting pathogen-specific regions.
- Screen for secondary structures such as hairpins and primer-dimers.
- Rank primer-probe combinations based on assay quality metrics.
- Export high-quality assay candidates for further experimental validation.

Methodology

The system will utilize python to analyze pathogen DNA sequences and generate candidate primers using a sliding-window approach. Primer candidates will be evaluated based on length, GC content, and melting temperature. Valid primer pairs producing amplicons between 70 and 200 bp will be selected, and corresponding TaqMan probes will be designed within the amplified region. Secondary structure analyses will identify potential hairpins and dimers, while a scoring algorithm will rank primer-probe sets according to their suitability for pathogen detection and quantification.

Expected Outcome

The project will produce an automated tool capable of generating optimized qPCR primer and probe sets for pathogen-specific detection assays. The system is expected to reduce the time and effort required for assay development while supporting rapid and reliable pathogen identification.

Significance of the Study

This project demonstrates the integration of bioinformatics, molecular biology, and software development in the field of molecular diagnostics. The resulting tool will provide researchers and students with a practical platform for automated qPCR assay design and contribute to efforts aimed at improving pathogen detection and disease surveillance through computational approaches.

Sources

MIQE 2.0: Revision of the Minimum Information for Publication of Quantitative Real-Time PCR Experiments Guidelines

Information on Rapid Molecular Assays, RT-PCR, and other Molecular Assays for Diagnosis of Influenza Virus Infection | Influenza (Flu) | CDC

Laboratory Techniques | IDPB | CDC

US CDC Real-Time Reverse Transcription PCR Panel for Detection of Severe Acute Respiratory Syndrome Coronavirus 2 - Volume 26, Number 8—August 2020 - Emerging Infectious Diseases journal - CDC

Timeline:

August 31- discussed roles and submitted project update 1

September 1-4 : Research and trying out pseudocode. Only research was completed on how host the app on GitHub and background information on project.

September 5-12: minors preparation and minor exams. Couldn't do much of the project.

September 15- 23: working on the pseudocode and redoing the code due to many errors.

September 24-27: refining the pseudocode with minimal errors and submitting project update 2